00-M-1243A
February 24, 1975
SUPERSEDING
Fed. Spec. 00-M-1243
December 14, 1972
(See 6.4)

FEDERAL SPECIFICATION

MOWER, LAWN, POWER

(HEAVY DUTY, INSTITUTIONAL, ROTARY, 21 INCH, GASOLINE ENGINE DRIVEN)

This specification was approved by the Commissioner,
Federal Supply Service, General Services Administration,
for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers 21-inch (nominal) rotary (blade rotating essentially in a horizontal plane), heavy duty, institutional, walk-behind mowers either hand-propelled or self-propelled with gasoline-engine power-driven cutting blades.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. The power lawn mowers covered by this specification shall be furnished in the following types and classes, as specified (see 6.2):

Type I - Hand-propelled.
  Class 1 - Four cycle.
  Class 2 - Two cycle.

Type II - Self-propelled.
  Class 1 - Four cycle.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-P-40 - Packaging and Packing of Hand Tools.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-L-2105 - Lubricating Oil, Gear, Multi-Purpose.
MIL-E-10062 - Engines: Preparation for Shipment and Storage of.
MIL-L-21260 - Lubricating Oil, Internal Combustion Engine Preservation and Break-In.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications: The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

American National Standards Institute (ANSI), Inc., Standard:

B71.1 - Safety Specification for Power Lawn Mowers.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

Metal Powder Industries Federation (MPIF), Standard:

35 - P/M Materials Standards and Specifications.

(Application for copies should be addressed to the Metal Powder Industries Federation, P.O. Box 2054, Princeton, NJ 08540.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P. Street, N.W., Washington, DC 20036.)

Society of Automotive Engineers (SAE) Standards:

J429 - Mechanical and Quality Requirements for Externally Threaded Fasteners.
J607 - Small Stock Two Stroke and Four Stroke Cycle Gasoline Engine Test Code.

(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, NY 10017.)

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Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Condition. Lawn mowers shall be complete and new. They shall be fully assembled except that handles need not be affixed to the machines. There shall be no evidence of damage to the mower or component parts that would appreciably affect safe usage or performance at the completion of any tests conducted to determine compliance with the requirements herein. Mowers shall be free from rust and imperfections which may affect appearance or serviceability of the mower.

3.1.1 Materials. The materials used in the construction of lawn mowers shall be of an adequate grade for the intended application.

3.2 Standard product. The lawn mowers furnished under this specification shall be the manufacturer's current standard industrial product except for those modifications or changes required in the standard product in order to comply with the requirements of this specification. All accessories and components shall be furnished with each lawn mower that are normally furnished with the standard product offered commercially.

3.3 Bid sample model. When specified (see 6.2), and prior to the award of the contract, the supplier shall furnish one mower for each bid to prove that his production methods and choice of design detail will produce lawn mowers that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the bid sample model during production shall be subject to the approval of the contracting officer. Approval of the bid sample model by the contracting agency shall not relieve the supplier of his obligation to furnish lawn mowers conforming to this specification. The bid sample model shall be retained (see 4.3) until the end of the contract for comparison examination.

3.4 Design and construction. The mowers shall be single cutting units and shall consist essentially of a gasoline engine, blade, housing, guards, controls, handle, wheels, and drive train as required. Unless otherwise specified (see 6.2), sizes of cut within a range of +2 inches to -1 inch of the nominal size specified will be accepted. All mowers shall conform to the requirements of ANSI B71.1- "Safety Specification for Power Lawn Mowers" as are applicable for walk-behind rotary lawn mowers. The mowers shall be capable of cutting in the hardiest of cultivated and uncultivated grasses and designed for trim cutting to within 1 inch of a vertical projection on at least one side of the housing (a vertical projection being defined as a wall). Lawn mowers serviced and maintained in accordance with manufacturers' instructions shall be capable of operating continuously 8 hours daily for 60 days under the cutting conditions and at the operating speeds specified by the manufacturer without the necessity for repairing or reconditioning the piston, piston rings, or cylinder bore or involving the repair to the differential or clutch. When specified (see 3.20 and 6.2), a grass catcher shall be furnished.

3.5 Engine. The engine shall be single cylinder of either the two cycle or four cycle type, as specified (see 1.2.1 and 6.2) and shall sufficiently power the lawn mower for heavy duty performance specified herein. It shall be equipped with a magneto ignition and a fuel strainer. Engine exhaust shall be directed below the horizontal plane or away from the operator zone. The starter shall be recoil type that automatically releases when the engine starts. The starter housing shall be mounted with removable threaded fasteners to facilitate disassembly. The engine rated horsepower specified below shall be maximum engine power rated at 3600 rpm per SAE J607 procedure. Maximum operating speed may be lower than 3600 rpm. The engine shall be designed and mounted so as to reduce the possibility of clogging the cooling and ventilation systems.

3.5.1 Class 1 - Four cycle engine. The four cycle engine shall be minimum 6 horsepower and shall have a crankcase of aluminum alloy die casting. The engine speed shall be controlled by mechanical type governor. The crankshaft housing shall be oil-tight and the shaft shall be supported (top and bottom) on ball or roller bearings or metal alloy plain bearing bushings. The engine shall have a float-type carburetor with a replaceable dry element or oiled polyurethane foam air cleaner. The lubrication system shall provide positive lubrication to the internal engine parts. Engines must operate on inclines of up to 26 percent (15°) without lubrication failure, (see 4.6.3.1). The crankcase oil capacity shall be a minimum of 25 ounces. Exhaust valves and seats shall be hard faced heat-resistant alloy. The crankshaft shall be not less than 25 mm (0.9843-inch) in diameter at the PTO extension and shall not decrease in diameter at the point of blade attachment. When a crankshaft adapter is used, it shall not increase the effective crankshaft length more than 1/2 inch.

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3.5.2 Class 2 - Two cycle engine. The two cycle engine shall be minimum 100 C.C. The crankcase shall be of aluminum alloy die casting and shall be scavenged. Cylinder liner shall be cast iron. Engine's crankshaft shall be not less than 19 mm (3/4-inch) in diameter at the PTO extension and shall be fitted with anti-friction bearings. The engine shall be equipped with a diaphragm or float-type carburetor, an oil-soaked polyurethane air cleaner and an air vane governor for controlling the engine speed. Engine lubrication shall be provided through the mixing of oil and gasoline. The oil grade and weight, gasoline/oil ratio and mixing instructions shall be included in the instruction booklet or pamphlet (see 3.22). The fuel system shall operate on inclines up to 100 percent (45°) (see 4.6.3.2).

3.5.3 Gas tank. The gas tank shall be steel or plastic and shall have a minimum capacity of one gallon. Wall thickness of a steel tank shall be minimum 19 gage. Wall thickness of a plastic tank shall be not less than 1/8-inch. If a plastic tank is furnished, the tank manufacturer shall certify that the formulation of the plastic in the tank is correct for storing gasoline, the manufacturing process is such as to produce a strong crack-resistant tank, the plastic has been treated with an ultraviolet inhibitor to prevent the degradation of the natural plastic resin and that the tanks are approved for their specific application for a minimum of five years life. The tank shall be securely mounted with metal fasteners and metal brackets. The mounting shall be secure enough so that the tank shall not vibrate loose under the mowing conditions and at the operating speeds specified herein. The filled tank and mounting shall be capable of withstanding the unbalance test of ANSI B71.1 without sustaining damage to welds, brackets, fasteners, mounting area or the tank. The method of fuel line attachment to the tank and carburetor shall insure positive hook up. The tank and fuel lines shall be mounted in a location that will prevent them from being snagged during mower operation.

3.5.4 Fungus proofing. When specified (see 6.2), the electrical circuitry of the engine shall be protected from the effect of water and fungus growth by an overall treatment, except as specified below, with a varnish conforming to MIL-V-173.

(a) Components which are inherently fungus and water-resistant or which are hermetically sealed need not be treated.
(b) Components whose function will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied to give a minimum dry film thickness of 0.001-inch to component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

3.5.5 Safety switch. When specified (see 6.2), engines shall be equipped with a high-temperature safety switch.

3.6 Blade. The blade shall be one-piece steel, designed for suction lift and balanced. The blade shall have a minimum 3-inch cutting edge at each end and shall be heat-treated to a hardness of Rockwell C 35 to 48 for the entire cutting edge length and blade width. Blade width shall be minimum 2 inches and thickness shall be minimum 3/16-inch. The engine shaft shall provide for blade mounting by means of a tapped threaded hole to receive a threaded fastener or by means of a threaded shaft locknut fastener. Fasteners shall be at least equal to SAE 5 grade. The blade shall be securely mounted to the engine shaft with a ferrous metal adapter. The blade shall not separate from the shaft in case of adapter failure.

3.7 Controls. All controls shall be of durable construction and designed for heavy duty use. The mowers shall have a control for regulating the engine speed and an engine shutoff device as required by ANSI B71.1. Type II mowers shall also have a wheel drive disengaging/engaging control. In addition to the requirements specified in ANSI B71.1, the engine speed control shall be mounted on the handle within the operator zone. All controls shall be positive in action and remain fixed, with the exception to deadman control application, in any desired position under usage conditions specified herein.

3.8 Handle assembly. The handle assembly shall be constructed of welded or seamless steel tubing or pipe with minimum 15-gage wall thickness and minimum 7/8-inch outside diameter. The design of a two-section handle shall include a cross-member which shall serve as reinforcement to increase the strength and rigidity. The handle assembly construction shall withstand the test in 4.6.2. The handle shall be adjustable to a minimum of three height positions. Each position shall have a positive stop to prevent movement of handle during normal operations. Handgrips shall be covered with rubber or other suitable industry accepted composition which shall not slip when twisted by hand.

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3.9 Blade housing. The blade housing shall consist of the mower deck, front, rear and side blade guards. The blade housing shall be steel, aluminum or magnesium alloy casting or combination aluminum-magnesium alloy casting. Steel housing shall have minimum 12-gage wall thickness. Alloy casting housing shall have minimum 6-gage wall thickness. Housing shall be reinforced at the wheel mounting and handle mounting points. If the front guard is not an integral part of the housing or is not permanently attached, a warning label shall be placed on the mower deck above the guard. The label shall indicate that it is dangerous to operate the mower without the guard or when it is positioned so that the bottom edge is higher than allowed for the front blade enclosure in ANSI B71.1. Reinforcement construction on the underside of the mower deck shall not form a trap for small rocks, gravel and other hard objects which could penetrate the engine housing when thrown against it by the blade rotation. The housing shall withstand the chassis impact test of 4.6.1 in addition to all applicable tests in ANSI B71.1.

3.10 Frame. If a steel blade housing is furnished (see 3.9) it shall be mounted on minimum 10-gage steel frame. Alternately, 12-gage steel frames with minimum 12-gage steel reinforcements sandwiched to the wheel mounting areas are acceptable. Frames shall give sufficient support to the wheels in all mowing operations, including operations on slopes of up to 26 percent (15°) for class 1 mowers and 100 percent (45°) for class 2 mowers. Frame structures shall withstand the same tests as required for blade housing.

3.11 Guards and shields. All guards and shields shall be provided as required by ANSI B71.1 for walk-behind rotary mowers.

3.12 Wheels and tires. There shall be four metal wheels of cast or disc-type construction. Discs shall be welded, bolted or riveted to each other. The wheels shall be fitted with semi-pneumatic treaded tires of minimum 8 inches diameter by 1.75 inches wide. Tread width of tires on type II mowers shall be minimum 1.75 inches. The wheels shall be equipped with ball or roller bearings. The bearings shall be sealed or a grease fitting shall be provided on each wheel spindle to facilitate adequate lubrication and cleansing to each bearing. Wheels shall be bolted directly to the frame or to reinforced areas of the blade housing. Bolts shall be heat-treated steel at least equal to SAE 5, and shall have a minimum shoulder diameter of 1/2-inch.

3.13 Cutting height. An adjustment shall be provided on each wheel to vary the cutting height throughout a minimum range of 1-1/2 inches to 3 inches above the ground with a minimum of four positions. The adjustments shall be positive and shall be held in position by bolt fasteners. Hand tools shall be required to adjust height positions.

3.14 Blade driving methods. Blade driving methods employing slipping clutches or other slipping or frictional devices shall not be acceptable. Set screws shall not be used either wholly or in part to transmit torque loads.

3.15 Weight. The weight of the type I mowers shall not exceed 96 pounds and the weight of the type II mowers shall not exceed 115 pounds. The weights designated are for unserviced mowers without optional grass catcher.

3.16 Fasteners. Bolt and nut threads shall conform to NBS H28. Swagging, peening, or staking of threads shall not be permitted. Sheet metal screws and sheet metal nuts shall not be used in structural applications unless approved by the procuring activity. Where aluminum-magnesium or zinc alloys are involved, assembly of parts shall be by the following: through bolts, bushings with tapped holes, threaded studs inserted to a depth of two stud diameters minimum, coil-type inserts, or machine screws in holes tapped directly in the metal where frequent disassembly is not required.

3.17 Lubrication. All components requiring lubrication shall be either permanently lubricated or shall be readily accessible for lubrication and provided with functional grease or oil fittings. Each fitting shall be suitable for the lubricant to be applied. Plugs provided in housings for lubrication purposes shall be metal screw type.

3.18 Finish. All exterior metal surfaces shall be finished with a smooth durable weatherproof paint or lacquer to resist corrosion. Handle frames may be chromium or nickel-plated.

3.19 Propulsion.

3.19.1 Type I. Type I mowers shall be propelled by hand pushing.

3.19.2 Type II. Type II mowers shall be self-propelled. The propulsion design shall be such that will drive the mower for continuous cutting of heavy grasses at a forward speed of 220 to 300 feet per minute on firm, fairly level courses and 175 feet per minute, minimum, directly up or across in both directions on firm, smooth grades of 26 percent (15° slope), (see 4.6.4).

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3.19.2.1 Drive method. The drive method shall be positive in action using either roller chain and sprocket, gear shaft and gears, V-belt, or a combination thereof to either the front or rear wheels. Mowers driving on the rear wheels shall be propelled through differential gearing or differential action over-running clutch, provided the sliding contact members are hardened to Rockwell C 50 minimum. All gears shall be suitable for continuous heavy duty use, shall mesh fully and smoothly, and shall be completely housed and properly lubricated. If a V-belt drive is supplied, the design shall provide for take up or slack to the V-belt. All pulleys shall be formed and designed for continuous heavy duty use and shall be securely mounted. Powder metallurgy steel pulleys, idlers and gears shall comply to MPIF Standard 35. V-belts used shall meet industry requirements for engine horsepower transmitted with a service factor of 1.5. At the time of bid submission, the bidder shall provide certification from the belt manufacturer for his propulsion design, that all belts utilized on the unit are approved for their specific application for a minimum life of 500 hours.

3.20 Grass catcher. When specified (see 6.2), a grass catcher shall be furnished with the mower. The catcher shall meet the requirements of ANSI B71.1 and shall be durable in service and be readily fastened to the lawn mower.

3.21 ANSI conformance. All mowers and grass catchers when furnished shall conform to all applicable requirements of ANSI B71.1. At the time of bid submission, the bidder shall furnish certification to the contracting officer that the specific model being offered has been tested and approved under the requirements of ANSI B71.1. Acceptable certification shall be a copy of the power equipment test report and approval issued and signed by an official of the independent testing laboratory which has the approval of Outdoor Power Equipment Institute (OPEI) for issuing certification labels. Each mower supplied to the Government shall display the OPEI Certification Label in a prominent location on the blade housing.

3.22 Instruction and parts list information. The contractor shall furnish with each lawn mower either a booklet(s) or pamphlet(s) giving complete instructions for the operation, lubrication, adjustment, and care of the engine, mower, and attachment units. A pamphlet lising all repair parts and parts' source information shall also be furnished with each lawn mower.

3.23 Identification marking. Each lawn mower shall be marked in a plain and permanent manner with the model designator and name of the manufacturer or with a trademark of such known character that the source of manufacture may be readily determined.

3.24 Workmanship. The mower shall be free from defects such as fractures, splits, punctures, tears, dents, creases, bows, miscasts, rust, deteriorations, or malformations. There shall be no sharp edges, burrs, or slivers which could cause injury to a person operating, handling, maintaining or repairing the mower.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Bid sample inspection. Bid sample inspection shall be applied to the mower submitted in accordance with 3.3. Failure of the bid sample model to meet all specification requirements and tests shall be cause for rejection. After acceptance of this model, the unit shall be retained by the contractor for use by the Government in the inspection of production lots. Design changes between the bid sample and production units must be approved by the Government. The bid sample model may be shipped in the last lot offered under the contract or, at the discretion of the contractor, this model may be retained for use in the event of additional contracts. In this event, the inspection and tests may be waived at the discretion of the Government.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with requirements of referenced subsidiary specifications and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.3.2 Inspection of the end item.

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4.3.2.1 Visual and dimensional examination. Completely assembled mowers shall be examined for defects listed in table I. The inspection level shall be level II of MIL-STD-105, with an acceptable quality level (AQL) of 4.0 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

4.3.2.2 Testing for quality conformance. The blade hardness test shall be performed at inspection level II with an AQL of 4.0 defects per hundred units. The contractor shall certify to the contracting officer that the model mower being furnished has passed all other applicable tests specified in 4.6. The tested model shall include all modifications made to a standard product which was required to bring it into compliance with the requirements of this specification. The certification shall cease to be valid when any component on the mower is changed or modified within the course of a contract. The government reserves the right to perform any tests necessary to determine mower compliance to this specification.

TABLE I. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not finished where required.	X	
	Type of finish not as specified, or blistered, peeled, chipped, or area of no film or thin film.	X	
	Any component that is painted or otherwise coated where serviceability may be affected.	X	
	Significant rust.	X	
	Negligible rust.		X
Dimensions	Dimensions not in accordance with the specified requirements.	X	
Weight	Exceeds maximum weight allowed.	X	
Type	Not type specified.	X	
Class	Not class specified.	X	
Construction and workmanship, general	Component missing or not as specified.	X	
	Malformed, fractured, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, or otherwise impaired.	X	
	Sharp burr, sliver, or splinter that may be injurious to personnel.	X	
	Component not readily accessible for servicing where required.		X
Welding or brazing (where required)	Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused.	X	
	Slag inclusion, undercut, not smooth and uniform, scale, or flux deposits not removed.		X

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Bolts, nuts, screws, studs, and other types of threaded fasteners.	Missing, broken, stripped, fractured, not as specified.	X	
	Loose		X
Assembly	Any component not properly assembled or secured.	X	
	Unit perceptibly out of square or alignment.	X	
Marking, identification, caution labels, certification labels, instruction labels, instruction and parts list manual.	Missing, incomplete, not legible, not as specified.	X	

4.6 Tests.

4.6.1 Chassis impact test. The mower shall be dropped 10 times to a concrete floor from a distance of 12 inches. The drop shall be such as to cause the machine to land essentially on all four wheels. Wheel alignment shall be measured before and after test to assure that no permanent deformation occurs during test. Acceptance shall require that none of the component parts shall fracture, break, loosen, or deform in a manner that would adversely affect performance or be hazardous to the operator.

4.6.2 Handle assembly test. The test shall consist of twisting the hand gripping bar clockwise, setting the mower on the two right wheels with the left wheels clearing the floor. The handle shall then be twisted counterclockwise until the right wheels clear the floor. This test shall be repeated 50 times and an inspection made. The handle position shall be measured before and after the test to verify that no permanent deformation occurs during the test. Any evidence of cracks or permanent deformation in the handle assembly or mounting area shall be cause for rejection.

4.6.3 Inclined operation tests. Mowers shall be tested for stability and the engine's ability to operate on inclined positions. The downhill wheels of the mower shall be blocked to prevent slippage. With the engine running beginning with a full tank of fuel, all four wheels of the mower shall remain in contact with the inclined surface. Failure of the mower engines to operate for the periods of time and in the positions specified below (see 4.6.3.1 and 4.6.3.2) or all wheels not remaining in contact with the surface shall be cause for rejection.

4.6.3.1 Class 1 - four cycle engine. The mower having a four cycle engine shall be positioned on a 15° inclined surface with the longitudinal axis of the mower parallel to the inclined edge (see figure 1). The engine shall run at approximately 3/4 throttle for two hours or until fuel is exhausted, whichever comes first. The test shall be repeated with the mower rotated and mounted 180° from the original position. Engines shall not stall, leak, excessively consume or burn oil.

4.6.3.2 Class 2 - two cycle engine. The two cycle engine test shall be performed with the mower positioned on an inclined surface of 45° in the two positions specified in 4.6.3.1 and in two additional positions, directly up and down the incline. The engine shall run at 3/4 throttle for a period of 15 minutes in each specified position prior to fuel starvation.

4.6.4 Propulsion capability test (type II mowers).

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4.6.4.1 Level ground. The propulsion capability test shall be conducted on a straight, firm, fairly level course. Whenever practicable, the course shall contain heavy, thick grass, or grass and weeds with a height of 4 to 5 inches. With the mower operating at the highest speed attainable and set at the 3-inch cutting height, it shall be guided toward a landmark or special marker while cutting grass for not less than 2 minutes to determine compliance with 3.19.2. The height adjustment shall then be set at the 1-1/2-inch setting and a second pass made over the same area to determine compliance with 3.19.2.

4.6.4.2 Grade. The grade propulsion test shall be conducted up and across, in both directions, on a grass course (see 4.6.4.1) with a 15° slope to determine compliance with 3.19.2. The course shall run between two marked points not less than 55 feet apart. With the mower operating at the highest speed attainable and set at the 1-1/2-inch cutting height, the time required (in seconds) for the mower to pass between the marked points shall be noted. The speed shall be calculated as follows:

$$\text{Speed (ft/minute)} = \frac{60 \times \text{distance (feet)}}{\text{time (seconds)}}$$

4.6.5 Blade hardness. The hardness of the blade shall be tested in not less than 3 areas on each end for compliance to 3.6.

4.7 Inspection of preparation for delivery. An inspection shall be made to determine that packaging, packing, and marking requirements are in compliance with section 5 of this specification. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 percent defectives.

TABLE II. Classification of preparation for delivery defects

Examine	Defect
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container, loose-strapping, inadequate stapling. Bulging or distortion of containers.
Preservation	Improperly applied or missing.
Containers	Not as specified, not close-fitting for packing level A or B.
Contents	Other than specified quantity per container.
Weight	Weight of contents exceeds container limitation.

5. PREPARATION FOR DELIVERY

5.1 Disassembly. Parts vulnerable to damage, pilferage, and loss or that would otherwise increase cubage, such as controls, control rods, levers, and handles shall be removed. All bolts, nuts, pins, and washers removed shall be enclosed in a suitable container (bag, sack or box) which shall be securely closed to prevent their loss.

5.2 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Unprotected metal surfaces. Unprotected metal surfaces requring the application of a contact preservative shall be coated with a type P-1 preservative conforming to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.1.2 Engine. The engine and engine accessories shall be preserved in accordance with MIL-E-10062, except that the exterior surfaces shall not be sprayed with insulation compound.

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5.2.1.3 Wheels. Wheels shall be removed from the mowers and the interior surfaces, gears,and drive control clutches coated with a lubricant conforming to MIL-L-2105 and the wheels replaced.

5.2.1.4 Drive chains and sprockets. Exposed drive chains shall be coated with enough type P-3 preservative to assure penetration to the inner surfaces of the rollers, pins, and bushings. After excess type P-3 preservative has drained, the entire chain shall be coated with type P-1 preservative. Sprockets and other unprotected surfaces of the drives shall be coated with type P-1 preservative.

5.2.1.5 Drive belts and pulleys. Pulley faces or grooves shall be coated with primer conforming to TT-P-664. The drive belts shall remain in place on the equipment and shall have the tension relieved.

5.2.1.6 Gears and gear housings. When the lubricant contained in the gear housing conforms to the specified operating lubricant, MIL-L-2105 or MIL-L-21260, a quantity of the applicable lubricant required to raise the lubricant to the operating level in the housing shall be added. The gears shall be actuated to insure coating of all interior parts and surfaces. The threads of each lubrication fill plug shall be coated with lubricant conforming to MIL-L-21260 before the plug is reinstalled. The gear housing may be filled by removing the top of the housing in lieu of through fill plugs. The top shall be secured in place by metal threaded fasteners.

5.2.1.7 Clutches. Metal components of clutches shall be coated with a covering suitable to protect them during overseas shipment or for controlled humidity storage.

5.2.1.8 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40.

5.2.1.9 Technical publications. Technical publications shall be preserved and packaged in accordance with MIL-P-116, method IC-1.

5.2.2 Level C. Mowers, parts, tools, and technical publications shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity for immediate use or protected, enclosed storage.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.3.1 Level A. One complete mower, parts, tools, and technical publications, preserved and packaged as specified in 5.2, shall be packed together in a close-fitting box conforming to PPP-B-621, class 2, style optional, or PPP-B-601, overseas type, style optional, or PPP-B-640, class 2, style optional. Blocking, bracing, or cushioning shall be provided to prevent the disassembled parts from coming in contact with each other. Closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.3.2 Level B. One complete mower, parts, tools, and technical publications preserved and packaged as specified in 5.2 shall be packed together as specified in 5.3.1, except that the boxes shall be class 1 domestic type. Alternatively, power lawn mowers shall be packed in a fiberboard box conforming to PPP-B-636, class domestic, variety DW, grade 350. The fiberboard box shall be fitted with four internal corner posts or a liner of fiberboard, grade not less than 275.

5.3.3 Level C. One complete mower shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the requirements of Uniform Freight Classification or National Freight Classification, as applicable to the mode of transportation.

5.4 Marking.

5.4.1 Civil agencies. In addition to any special markings required by the contract or purchase order (see 6.2), shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military agencies. In addition to any special markings required by the contract or purchase order (see 6.2), shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The mowers covered by this specification are intended for heavy duty commercial use in the care and maintenance of lawns and grounds for which the Government is responsible.

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6.2 Ordering data. Purchasers shall select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class (see 1.2.1).
(c) When bid sample is required (see 3.3).
(d) When grass catcher is required (see 3.4 and 3.20).
(e) Width of cut, if different (see 3.4).
(f) Fungus proofing, if required (see 3.5.4).
(g) High temperature safety switch, if required (see 3.5.5).
(h) Levels of preservation, packaging, and packing required (see 5.2 and 5.3).
(i) Special marking, if required (see 5.4).

6.3 Service guarantee. It is suggested that the purchasing officer include the following clause in the invitation for bids and all awarded contracts:

All necessary adjustments of machines, with the exception of engines which are subject to the warranties of their respective manufacturers, procured hereunder, not occasioned by misuse or accident through fault or negligence by the Government, shall be made by the contractor at his own expense, including transportation costs, if any, during the 90-day period after the machine is placed into service. All mowers procured hereunder shall be guaranteed for a period of six months after the date the machine is placed into service. During the guarantee period, all broken or defective parts not caused by accident or misuse through fault or negligence by the Government, must be replaced (including labor, transportation, and parts) at the contractor's expense. Also, all necessary machine adjustments occasioned by such defective parts must be made at the contractor's expense, including transportation costs, if any. The guarantee shall also apply to breakage of mower parts caused by objects thrown by the cutting knives. The contractor shall agree to have all parts for the item furnished available for a period of five years from the date of award, and shall agree to bill the Government for spare parts not covered by the warranty at a price not to exceed their industry-wide published list price. The contractor shall agree that during this five year period of availability in-stock parts will be shipped immediately and out-of-stock parts will be shipped within 30 days after receipt of order. The contractor shall notify the ordering office of the expected shipping date of out-of-stock parts.

6.4 Supersession data. When finalized, this specification is intended to supersede type II, class 2, style 1, sizes A and B and type II, class 2, style 2, sizes A and B of 00-M-00681c(GSA-FSS), dated April 30, 1965, and type II, styles 1 and 2, size A of 00-M-681a, dated August 30, 1955.

6.5 Paragraphs in this revision have not been marked to indicate where changes from the previous issue were made due to the extensive changes throughout the specification.

MILITARY CUSTODIANS:

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MILITARY COORDINATING ACTIVITY:

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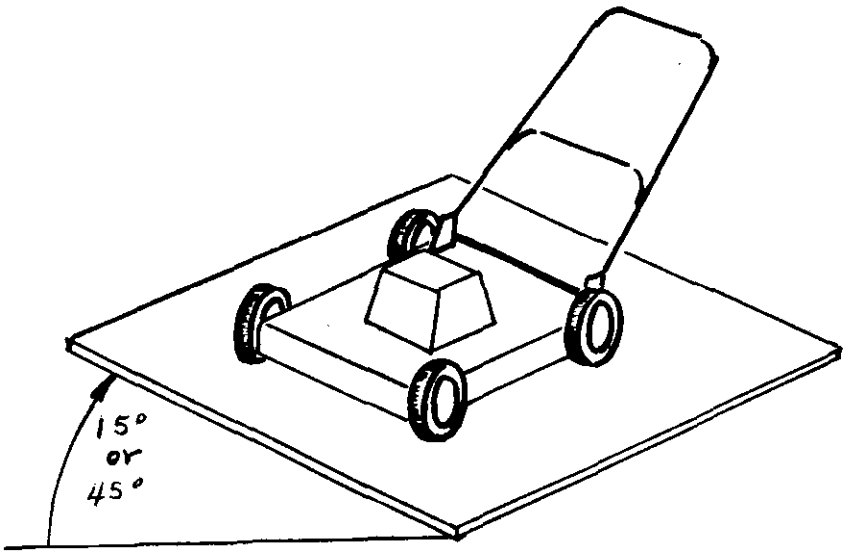


FIGURE 1.

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